

GIS: Geography: Information Science Courses

Courses

GIS 3905 Directed Study

College of Sci and Engineering, Department of Earth & Environmental Sciences

1-12 sh (may be repeated indefinitely for credit)

GIS 4006 Computer Cartography

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GIS 4043/L

Co-requisite: GIS 4006L

The course focuses on the fundamentals of cartography, spatial statistics, thematic mapping techniques, and web-based mapping. Students will gain an interdisciplinary understanding of cartographic representation and visualization with hands-on applications using cutting-edge GIS software to create purpose-tailored maps. Upon successful completion of this course, students will be able to interpret and appropriately communicate spatial data; will have developed a personalized cartographic style; will have created a professional GIS portfolio for current/potential employers; and most importantly will have developed a keen appreciation for maps and spatial awareness. Offered concurrently with GIS 5007 where graduate students will be assigned additional work.

GIS 4006L Computer Cartography Lab

College of Sci and Engineering, Department of Earth & Environmental Sciences

1 sh (may not be repeated for credit)

Prerequisite: GIS 4043/L

Co-requisite: GIS 4006

The course focuses on the fundamentals of cartography, spatial statistics, thematic mapping techniques, and web-based mapping. Students will gain an interdisciplinary understanding of cartographic representation and visualization with hands-on applications using cutting-edge GIS software to create purpose-tailored maps. Upon successful completion of this course, students will be able to interpret and appropriately communicate spatial data; will have developed a personalized cartographic style; will have created a professional GIS portfolio for current/potential employers; and most importantly will have developed a keen appreciation for maps and spatial awareness. Prior to enrollment, students should be competent with Windows operating system including storage, copying and management of multiple data types, managing multiple windows and applications, and disciplined to save work frequently; basic competence with ArcGIS Pro and Microsoft Excel. Offered concurrently with GIS 5007L where graduate students will be assigned additional work.

GIS 4035 Photo Interpretation and Remote Sensing

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GIS 4043*/L*

This course is designed to familiarize students with the fundamentals of remote sensing and photo interpretation through hands-on techniques with aerial photographs and satellite imagery based on real-world applications. Both active and passive sensors will be discussed in lecture. The course is broken up into two distinct sections – interpreting digital aerial photographs and examining and manipulating digital data from satellites and other remote sensors. The lecture will focus on the history, technology, concepts, processes, and applications of Aerial Photography and Remote Sensing. Offered concurrently with GIS 5027, where graduate students will be assigned additional work.

GIS 4035L Photo Interpretation and Remote Sensing Lab

College of Sci and Engineering, Department of Earth & Environmental Sciences

1 sh (may not be repeated for credit)

Prerequisite: GIS 4035*

This course is designed to familiarize students with the fundamentals of remote sensing and photo interpretation through hands-on techniques with aerial photographs and satellite imagery based on real-world applications. Both active and passive sensors will be discussed in lecture. The course is broken up into two distinct sections – interpreting digital aerial photographs and examining and manipulating digital data from satellites and other remote sensors. The lab will focus on techniques for the practical use of digital aerial photography and satellite imagery using both Erdas Imagine and ArcGIS Pro. Equipment fees will be assessed. Offered concurrently with GIS 5027L where graduate students will be assigned additional work. Permission is required. Basic competency with ArcGIS Pro software is required. Prior coursework including Introduction to GIS (GIS 4043/GIS 5050 with lab) is expected.

GIS 4043 Geographic Information Systems

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Co-requisite: GIS 4043L

This introductory course strives to provide a comprehensive understanding of Geographic Information Systems (GIS) by striking a balance between the "how-to" of GIS using ArcGIS Pro software and the "why" of Geographic Information Science. Through exploration and analysis of local, regional, and global problems, students will gain valuable insights into the world of geospatial science and technology. Explore the fundamental concepts of representing real-world geographic data in a digital format, setting the foundation for geospatial analysis. Construct spatial databases, including data modeling and data organization techniques, ensuring data integrity and efficiency. Learn the power of spatial database queries enabling the extraction of valuable insights. Gain proficiency in spatial analysis techniques, using GIS tools to answer complex spatial questions and solve real-world problems. Create visually compelling and informative maps and visualizations through the basic principles of cartographic design. Explore the wide-ranging applications of geospatial technology across various disciplines for both professional and research uses. The lecture component of the course provides students with the theoretical underpinnings of GIS, helping them grasp the core concepts and principles underlying geospatial technology. This course provides a dynamic and hands-on learning experience, equipping students with the knowledge and skills needed to excel in the field of GIS. Whether you are interested in environmental science, urban planning, or any other discipline that relies on spatial data, this course is your gateway to unlocking the potential of Geographic Information Systems. Offered concurrently with GIS 5050 where graduate students will be assigned additional work. Permission is required. Credit cannot be received for both GIS 4043 and GEO 4151.

GIS 4043L GIS Laboratory

College of Sci and Engineering, Department of Earth & Environmental Sciences

1 sh (may not be repeated for credit)

Co-requisite: GIS 4043

This hands-on lab course is designed to complement the theoretical foundations covered in the GIS lecture course. It provides students with a practical and immersive experience in Geographic Information Systems (GIS), emphasizing fundamental concepts and techniques. By the end of this course, students will have gained the knowledge and skills necessary to apply GIS effectively in various professional and academic settings as well as developed a strong foundation for GIS project development, execution, and presentation. Equipment fees will be assessed. Offered concurrently with GIS 5050L where graduate students will be assigned additional work. Permission is required.

GIS 4048 Applications in Geographic Information Systems

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GIS 4043/L

The Applications in GIS course strives to provide a balance between the "how-to" of using ArcGIS Pro and the "why" of GIS by explaining the roles GIS technology plays in analyzing local, regional, and international problems. The course builds upon topics covered in Introduction to GIS (GIS 4043). Major components of the course include computer representation of geographic information, the construction of GIS databases, spatial analysis with GIS, application areas of GIS, and social and management issues that concern GIS. Applications in GIS will introduce the applied use of GIS (systems and science) in the study of geohazards, natural disasters, urban planning, homeland security/law enforcement, and marketing/location decisions. The course is directed at giving students an understanding of and experience with the practical uses of GIS software and data in these applications. Laboratory exercises, case studies, and course projects will use true-to-life datasets to solve real-world problems. The course will also focus on the development, execution, and presentation of GIS maps and materials. Since the visual communication of quantitative data is a vital skill for most GIS industries, this course will help you further develop your GIS skills by framing an issue, developing a set of high-quality GIS maps to illustrate the issue, and presenting a focused summary of your methodology and findings to your colleagues. Equipment fees will be assessed. Offered concurrently with GIS 5100 where graduate students will be assigned additional work.

GIS 4071 Methods and Techniques in Environmental Resource Management and Planning

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GEO 3210/L

Tools, methods, and techniques employed in the study of environmental impact and resource management. Research fundamentals studied and applied to environmental problems such as land use, environmental impact studies, Florida's development of regional impact, resource evaluation, and other topics. Material and supply fees will be assessed.

GIS 4102 GIS Programming

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GIS 4043/L

In today's technology driven world we are fortunate to have access to many tools with which to analyze and explore digital spatial data. During this course students will learn to use programming techniques to create applications that perform fundamental spatial analysis and automation tasks, such as geoprocessing, editing, database management, projecting data, and map creation. The course will focus primarily on using the Python programming language within the context of Esri's desktop GIS environment. Equipment fees will be assessed. Offered as a stand-alone course online and concurrently with GIS 5103, where graduate students will be assigned additional work. Permission required. Credit may not be received in both GIS 4102 and GIS 5103.

GIS 4301 GIS for Environmental Analysis

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GIS 4043/L

The course examines the use of geographic information systems (GIS) for environmental analysis. Students will use ESRI ArcGIS Pro to learn computer-based spatial analysis for conducting real-world environmental analysis using real data. The course builds upon the topics covered in Introduction to GIS (GIS 4043) but goes beyond the basic concepts and theories. The course will introduce the applied use of GIS to analyze geohazards, natural disasters, urban planning, hydrology modeling, and landform/landscape pattern analysis to name a few. Offered concurrently with GIS 5305. Graduate students will be assigned additional work.

GIS 4905 Directed Study

College of Sci and Engineering, Department of Earth & Environmental Sciences

1-12 sh (may be repeated indefinitely for credit)

GIS 4930 Special Topics in Geographic Information Science

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may be repeated for up to 6 sh of credit)

Prerequisite: GIS 4043/L

Today, geographic information systems (GIS) are used by everyone from scientists to everyday citizens to solve geographic problems ranging from the very simple to the highly complex. As the use of GIS and the availability of digital data increase, GIS users need to be aware of how the data being put into a GIS affects the reliability of the information products produced. Producing new and valuable information from spatial data requires a thorough understanding of their limitations and the methods used to process them. Students explore GIS theory and practice related to spatial data visualization, measurement, transformation, and optimization. An underlying theme that uncertainty is an inherent characteristic of spatial data is thoroughly examined, and students learn how to identify, measure, and live with it. By the end of this course, students will have gained extensive knowledge about various GIS analysis techniques, methods, outputs, and uncertainties related to specific problems experienced by many cities (large and small) worldwide. Combining lessons learned in previous GIS courses with more in-depth techniques presented in this course allows students to piece together previous knowledge and understand what it means to conduct GIScience. Equipment fees will be assessed. Offered concurrently with GIS 5935, where graduate students will be assigned additional work. Permission is required. Proficiency with ArcGIS Pro software is required. Prior coursework including GIS 4043/GIS 5050 with lab, GIS 4006/GIS 5007 with lab, GIS 4102/ GIS 5103, and GIS 4048/GIS 5100 is expected.

GIS 4944 GIS Internship

College of Sci and Engineering, Department of Earth & Environmental Sciences

1-3 sh (may not be repeated for credit)

Prerequisite: GIS 4048

GIS Internship is a supervised application of Geographic Information Science (GIS) in business, government, non-profit, educational, or other related organizations. The course has two major goals; to monitor the progress of students in their selected internships and to prepare students for the GIS workforce. This is one of the last courses required in the GIS Certificate and GIS Minor. Students completing a traditional internship (130 hours in the workplace) are classified as Group 1. Internships may be conducted on-site, remotely, or a mixture of the two. Up to 22 hours of experience completed through the seminar portion of the course may be contributed to internship hours if needed. Students unable to secure an internship will be classified as Group 2 and complete specified ESRI Training online courses totaling 130 hours. This course is taught completely online. The Canvas course site is designed to monitor students as they progress in their GIS Internship and provides activities and mechanisms that aid GIS career development. Seminar assignments include readings focusing on specific industries and related resources, recorded industry spotlight interviews, class discussions, and assignments that prepare students for the workforce. At the end of the course, students complete a GIS portfolio showcasing skills, best works throughout the GIS Certificate program or GIS minor courses, and a vetted resume. Offered concurrently with GIS 5945 where graduate students will be assigned additional work. Permission is required.

GIS 5007 Computer Cartography

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GIS 5050/L

Co-requisite: GIS 5007L

The course focuses on the fundamentals of cartography, spatial statistics, thematic mapping techniques, and web-based mapping. Students will gain an interdisciplinary understanding of cartographic representation and visualization with hands-on applications using cutting-edge GIS software to create purpose-tailored maps. Upon successful completion of this course, students will be able to interpret and appropriately communicate spatial data; will have developed a personalized cartographic style; will have created a professional GIS portfolio for current/potential employers; and most importantly will have developed a keen appreciation for maps and spatial awareness. Offered concurrently with GIS 4006 where graduate students will be assigned additional work.

GIS 5007L Computer Cartography Lab

College of Sci and Engineering, Department of Earth & Environmental Sciences

1 sh (may not be repeated for credit)

Prerequisite: GIS 5050/L

Co-requisite: GIS 5007

The course focuses on the fundamentals of cartography, spatial statistics, thematic mapping techniques, and web-based mapping. Students will gain an interdisciplinary understanding of cartographic representation and visualization with hands-on applications using cutting-edge GIS software to create purpose-tailored maps. Upon successful completion of this course, students will be able to interpret and appropriately communicate spatial data; will have developed a personalized cartographic style; will have created a professional GIS portfolio for current/potential employers; and most importantly will have developed a keen appreciation for maps and spatial awareness. Prior to enrollment, students should be competent with the Windows operating system including storage, copying and management of multiple data types, managing multiple windows and applications, and discipline to save work frequently; basic competence with ArcGIS Pro and Microsoft Excel is recommended. Offered concurrently with GIS 4006L where graduate students will be assigned additional work.

GIS 5027 Aerial Photography and Remote Sensing

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GIS 5050/L

This course is designed to familiarize students with the fundamentals of remote sensing and photo interpretation through hands-on techniques with aerial photographs and satellite imagery based on real-world applications. Both active and passive sensors will be discussed in lecture. The course is broken up into two distinct sections – interpreting digital aerial photographs and examining and manipulating digital data from satellites and other remote sensors. The lecture will focus on the history, technology, concepts, processes, and applications of Aerial Photography and Remote Sensing. Offered concurrently with GIS 4035 where graduate students will be assigned additional work. Permission is required. Basic competency with ArcGIS Pro software is required. Prior coursework including Introduction to GIS (GIS 4043/GIS 5050 with lab) is expected.

GIS 5027L Aerial Photography and Remote Sensing Lab

College of Sci and Engineering, Department of Earth & Environmental Sciences

1 sh (may not be repeated for credit)

Prerequisite: GIS 5050/L

Co-requisite: GIS 5027

This course is designed to familiarize students with the fundamentals of remote sensing and photo interpretation through hands-on techniques with aerial photographs and satellite imagery based on real-world applications. Both active and passive sensors will be discussed in lecture. The course is broken up into two distinct sections – interpreting digital aerial photographs and examining and manipulating digital data from satellites and other remote sensors. The lab will focus on techniques for the practical use of digital aerial photography and satellite imagery using both Erdas Imagine and ArcGIS Pro. Offered concurrently with GIS 4035L where graduate students will be assigned additional work.

GIS 5050 Geographic Information Systems

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Co-requisite: GIS 5050L

This introductory course strives to provide a comprehensive understanding of Geographic Information Systems (GIS) by striking a balance between the "how-to" of GIS using ArcGIS Pro software and the "why" of Geographic Information Science. Through exploration and analysis of local, regional, and global problems, students will gain valuable insights into the world of geospatial science and technology. Explore the fundamental concepts of representing real-world geographic data in a digital format, setting the foundation for geospatial analysis. Construct spatial databases, including data modeling and data organization techniques, ensuring data integrity and efficiency. Learn the power of spatial database queries enabling the extraction of valuable insights. Gain proficiency in spatial analysis techniques, using GIS tools to answer complex spatial questions and solve real-world problems. Create visually compelling and informative maps and visualizations through the basic principles of cartographic design. Explore the wide-ranging applications of geospatial technology across various disciplines for both professional and research uses. The lecture component of the course provides students with the theoretical underpinnings of GIS, helping them grasp the core concepts and principles underlying geospatial technology. This course provides a dynamic and hands-on learning experience, equipping students with the knowledge and skills needed to excel in the field of GIS. Whether you are interested in environmental science, urban planning, or any other discipline that relies on spatial data, this course is your gateway to unlocking the potential of Geographic Information Systems. Offered concurrently with GIS 4043 where graduate students will be assigned additional work. Permission is required.

GIS 5050L Geographic Information Systems Lab

College of Sci and Engineering, Department of Earth & Environmental Sciences

1 sh (may not be repeated for credit)

Prerequisite: GIS 5050*

Co-requisite: GIS 5050

This hands-on lab course is designed to complement the theoretical foundations covered in the GIS lecture course. It provides students with a practical and immersive experience in Geographic Information Systems (GIS), emphasizing fundamental concepts and techniques. By the end of this course, students will have gained the knowledge and skills necessary to apply GIS effectively in various professional and academic settings as well as developed a strong foundation for GIS project development, execution, and presentation. Offered concurrently with GIS 4043L where graduate students will be assigned additional work. Permission is required.

GIS 5100 Applications in Geographic Information Systems

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GIS 5050/L

The Applications in GIS course strives to provide a balance between the "how-to" of using ArcGIS Pro and the "why" of GIS by explaining the roles GIS technology plays in analyzing local, regional, and international problems. The course builds upon topics covered in Introduction to GIS (GIS 4043 / GIS 5050). Major components of the course include computer representation of geographic information, the construction of GIS databases, spatial analysis with GIS, application areas of GIS, and social and management issues that concern GIS. Applications in GIS will introduce the applied use of GIS (systems and science) in the study of geohazards, natural disasters, urban planning, homeland security/law enforcement, and marketing/location decisions. The course is directed at giving students an understanding of and experience with the practical uses of GIS software and data in these applications. Laboratory exercises, case studies, and course projects will use true-to-life datasets to solve real-world problems. The course will also focus on the development, execution, and presentation of GIS maps and materials. Since the visual communication of quantitative data is a vital skill for most GIS industries, this course will help you further develop your GIS skills by framing an issue, developing a set of high-quality GIS maps to illustrate the issue, and presenting a focused summary of your methodology and findings to your colleagues. Equipment fees will be assessed. Offered concurrently with GIS 4048; graduate students will be assigned additional work.

GIS 5103 GIS Programming

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

In today's technology driven world we are fortunate to have access to many tools with which to analyze and explore digital spatial data. During this course students will learn to use programming techniques to create applications that perform fundamental spatial analysis and automation tasks, such as geoprocessing, editing, database management, projecting data, and map creation. The course will focus primarily on using the Python programming language within the context of Esri's desktop GIS environment. Offered as a stand-alone course online and concurrently with GIS 4102 where graduate students will be assigned additional work. Permission is required. Equipment fees will be assessed. Credit may not be received in both GIS 5103 and GIS 4102. Basic competency with ArcGIS software is required. Completion of GIS 4043 with Lab or GIS 5050 with lab is recommended.

GIS 5305 GIS for Environmental Analysis

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GIS 5050/L

The course examines the use of geographic information systems (GIS) for environmental analysis. Students will use ESRI ArcGIS Pro to learn computer-based spatial analysis for conducting real-world environmental analysis using real data. The course builds upon the topics covered in Introduction to GIS (GIS 5050) but goes beyond the basic concepts and theories. The course will introduce the applied use of GIS to analyze geohazards, natural disasters, urban planning, hydrology modeling, and landform/landscape pattern analysis to name a few. Offered concurrently with GIS 4301. Graduate students will be assigned additional work.

GIS 5935 Special Topics in Geographic Science

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may be repeated for up to 6 sh of credit)

Prerequisite: GIS 5007/L AND GIS 5050/L AND GIS 5100 AND GIS 5103

Today, geographic information systems (GIS) are used by everyone from scientists to everyday citizens to solve geographic problems ranging from the very simple to the highly complex. As the use of GIS and the availability of digital data increase, GIS users need to be aware of how the data being put into a GIS affects the reliability of the information products produced. Producing new and valuable information from spatial data requires a thorough understanding of their limitations and the methods used to process them. Students explore GIS theory and practice related to spatial data visualization, measurement, transformation, and optimization. An underlying theme that uncertainty is an inherent characteristic of spatial data is thoroughly examined, and students learn how to identify, measure, and live with it. Equipment fees will be assessed. By the end of this course, students will have gained extensive knowledge about various GIS analysis techniques, methods, outputs, and uncertainties related to specific problems experienced by many cities (large and small) worldwide. Combining lessons learned in previous GIS courses with more in-depth techniques presented in this course allows students to piece together previous knowledge and understand what it means to conduct GIScience. Offered concurrently with GIS 4930 where graduate students will be assigned additional work. Permission is required. Proficiency with ArcGIS Pro software is required. Prior coursework including GIS 4043/GIS 5050 with lab, GIS 4006/GIS 5007 with lab, GIS 4102/GIS 5103, and GIS 4048/GIS 5100 are expected.

GIS 5945 GIS Internship

College of Sci and Engineering, Department of Earth & Environmental Sciences

1-3 sh (may not be repeated for credit)

Prerequisite: GIS 5100

GIS Internship is a supervised application of Geographic Information Science (GIS) in business, government, non-profit, educational, or other related organizations. The course has two major goals; to monitor the progress of students in their selected internships and to prepare students for the GIS workforce. This is one of the last courses required in the Graduate GIS Certificate program. Students completing a traditional internship (130 hours in the workplace) are classified as Group 1. Internships may be conducted on-site, remotely, or a mixture of the two. Up to 22 hours of experience completed through the seminar portion of the course may be contributed to internship hours if needed. Students unable to secure an internship will be classified as Group 2 and complete specified ESRI Training online courses totaling 130 hours. This course is taught completely online. The Canvas course site is designed to monitor students as they progress in their GIS Internship and provides activities and mechanisms that aid GIS career development. Seminar assignments include readings focusing on specific industries and related resources, recorded industry spotlight interviews, class discussions, and assignments that prepare students for the workforce. At the end of the course, students complete a GIS portfolio showcasing skills, best works throughout the GIS Certificate program or GIS minor courses, and a vetted resume. Offered concurrently with GIS 4944 where graduate students will be assigned additional work. Permission is required.

GIS 6005 Communicating GIS

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

This course begins with the basic theory of graphic design, cartography, and map production and distribution. Students then learn to communicate specific types of spatial and analytical information through maps, written and oral explanations, graphs, tables, charts, and interactive web mapping applications. Course includes lecture, hands-on exercises, written reports, and a final presentation. Restricted to students majoring in MS GIS Administration program, it is best taken during one of the first three semesters. Basic competency in GIS concepts and software is expected.

GIS 6105 Spatial Data Management

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

This course begins with the basic theory of database design. It then proceeds on to incorporate spatial data and its unique data management requirements. Students then learn how to extract, transform and load spatial data and its associated attribute data using specific GIS case study workflows. Course includes lecture, hands-on exercises, written reports, and a final project with a presentation requirement. Prior coursework in Introduction to GIS and GIS Programming is recommended.

GIS 6110 Advanced Topics in Geographic Information Science

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

This class brings together a number of open-source GIS tools in order to educate students on the methods and processes behind web-based geographic information systems (Web GIS). Specifically, we will review techniques with spatial databases, web mapping application programming interfaces (APIs), geospatial scripting, and theories of map interaction within the context of Web GIS. In order to achieve this near-complete picture of how Web GIS works use a pattern of software architecture known as model-view-controller or MVC. Finally, we will cover design considerations for Web GIS within browsers on mobile devices. Credit cannot be received for both GIS 6110 and GEO 6159. Competency with the principles of GIS and ArcGIS Pro software is required. Prior upper-level GIS coursework (GIS 5100, GIS 5935, GIS 5103) is recommended.

GIS 6555 Geographic Information Systems Management

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may not be repeated for credit)

Prerequisite: GIS 5935

This course provides practical information on the development, implementation, and operation of GIS programs and projects intended for both seasoned and aspiring GIS managers. The course focuses on planning and implementing GIS solutions for government agencies and contractors. The course combines lecture, discussion, and group exercises. An end-of-term project involves writing in response to real or hypothetical solicitations for a project that targets GIS tool development, implementation, and/or training to support management activities in local, regional, state, national, or international contexts. Restricted to students in the MS GIS Administration program and designed to be the last core GIS course completed before enrolling in GIS 6955 GIS Capstone.

GIS 6905 Directed Study

College of Sci and Engineering, Department of Earth & Environmental Sciences

1-12 sh (may be repeated indefinitely for credit)

GIS 6955 GIS Capstone

College of Sci and Engineering, Department of Earth & Environmental Sciences

3 sh (may be repeated for up to 6 sh of credit)

Prerequisite: GIS 6005 AND GIS 6110 AND GIS 6555

A final capstone experience for students who are nearing completion of their MS GIS Administration program. Prior to enrollment, students must submit a capstone intention form to program faculty outlining their project idea, relevant research, and potential client(s). Upon faculty and departmental approval, the student may enroll in capstone and begin writing a draft project proposal, including a literature review for committee review. By the end of the first semester, students are expected to have completed a final proposal and gain approval from their review committee before enrolling in the second capstone course. The second semester of the capstone is dedicated to carrying out the project and presenting findings and/or products in the form of (at least) a paper and presentation. Students work in collaboration with local partners, faculty, or the student's current employer to develop a real-world GIS application. Working independently, students: communicate with project partners to identify project goals; acquire and prepare spatial data for GIS data analysis; communicate with project partners to assess progress; manage spatial data; and produce necessary outputs for presentation as part of a final report. This final project should affirm the student's ability to think critically and creatively, to solve practical problems, to make reasoned and ethical decisions, and to communicate effectively. The capstone course serves as documentation of the student's personal mastery of professional competencies. It is designed to be an integrative experience for MS GIS Administration students. The course is restricted to students in their last two semesters of the MS GIS Administration program and upon completion of all previous GIS Core program course requirements.

* This course may be taken prior to or during the same term.